

Course Introduction

COSC 304 – Introduction to Database Systems



Introductions – Dr. Ramon Lawrence

Professor of Computer Science and Department Head

Research area: database systems, AI in education

Teaching experience:

- 2020 Killam Teaching Prize (top teaching award at UBCO – one per year)
- 2017 UBCO Teaching Excellence Award Winner (two per year)
- 9-time member of teaching honour roll (top 10% instructors)

Industry experience: GE Big Data, UnityJDBC company/consulting

Note: May address me as "Dr. Lawrence", "Professor", or "Ramon" (pronounced RAY-MUN).

The Essence of the Course

The overall goal of this course is for you to:

Create, query, design, and program with databases to build applications and perform data analysis.

The course covers relational and NoSQL databases, SQL, database design, database programming, JSON, and XML.

This is an important, **required** course as databases are fundamental to most modern information systems. SQL fluency is an important skill for developers and data analysts.

My Course Goals

- 1) Provide the information in an effective way for learning.
- 2) Inspire and motivate students to learn and appreciate the benefits of the course.
- 3) Strive for **all** students to understand the material and excel.
- 4) Be available for questions during scheduled times, office hours, and at other times as needed.
- 5) Teach students how to be a sophisticated database user (by understanding SQL), a database application programmer, and a database designer.
- 6) *Demonstrate and integrate AI techniques with database systems.***

Course Objectives

- 1) Understand the use case for databases and the relational model for data storage.
- 2) Fluency in SQL including schema definition (CREATE, DROP), data modification (INSERT, UPDATE, DELETE), and queries using SELECT.
- 3) Read existing database designs, design new databases using ER/UML modeling, and convert to the relational model.
- 4) Construct programs that access a database to read data, perform analysis, and output results.
- 5) Exposure to additional database technologies and data representations including NoSQL databases, JSON, XML, and database hosting.
- 6) Use AI to assist with SQL and database code generation and critically evaluate AI outputs.

Academic Integrity

Labs and Assignments

- Submit work completed by you and your partner.
- AI tools such as ChatGPT and GitHub Copilot may be used unless otherwise specified. You are responsible for understanding, critically evaluating, and being able to explain all submitted work.
- Do not share or receive assignment solutions outside your group.

Exams

- Exams are individual work. Use only materials explicitly permitted by the instructor.
- AI, Internet resources, and assistance from other people are not permitted.

Academic misconduct may result in serious academic consequences.
See the course syllabus for the complete policy.

How to Excel in This Course

Attend **every** class:

- Read notes **before** class as preparation and complete the questions.
- Participate in class exercises and questions.

Complete all lab assignments:

- Labs practice the fundamental employable skills as well as being for marks.

Practice on your own. Practice makes perfect.

- Do more questions than in the labs. AI can generate practice questions.
- Read the additional reference material.

The Database Projects

Two database project paths for different skill sets and interests:

- **Development Stream:** Build a database-backed web application
- **Analysis Stream:** Perform data analysis, reporting, and visualization

Key skill sets used:

- Database/web programming, database design using ER/UML
- SQL, Excel, visualization and reporting
- AI-assisted database development and analysis

Features:

- Project is integrated in the assignments and is a significant part of the mark.
- Students select one stream to do for marks but have access to all materials for both streams. Project assignments start at lab 7.
- Project assignments are done in groups of **two**.

The Lab Assignments

The lab assignments are critical to learning the material and are designed to build up your skills and prepare you for the exams.

The assignments are worth **20%** of your overall grade.

- Labs have an expected completion week and are normally due Friday of the following week.
- 10% bonus for completion by the Monday following the lab week; 5% bonus by Wednesday.
- Bonus requires demonstrating the lab during a help session.
- Lab assignments may take between 2 and 10 hours depending on the lab.

Assignments are done in **pairs**. No late assignments will be accepted.

There is no scheduled lab time. TA help desk hours are posted.

Quiz and Practice Questions

Quizzes and practice questions are worth 10% of the course grade.

- Points are earned through Canvas quizzes (asynchronous), iClicker questions, and PrairieLearn live questions (synchronous).
- $(\text{Canvas quiz mark} + \text{Clicker Mark} + \text{Live questions}) / 250 * 10\%$.
- Maximum is 10%.

More opportunities are provided than are required to earn the full 10%, so missing an occasional activity will not reduce your grade.

There are no make-ups for missed questions.

Systems and Tools

Course materials and code examples are hosted on GitHub.

Quizzes, discussion forums, and videos are on Canvas.

The assignments use Docker to deploy databases on your machine. A laptop is required for development.

All software used will be open source or free to install on your computer. You do not need to access hardware/software at UBCO.

Get help at <https://coursehelp.ubc.ca>.

My Expectations

My goal is for you to learn the material and walk out of this course confident in your abilities to:

- Query and update databases
- Write programs and web applications that interact with databases
- Design and model relational databases using ER/UML
- Use AI effectively while critically evaluating its output

I have high standards on the amount and difficulty of material that we cover. I expect a strong, continual effort in keeping up with readings, doing assignments, and working on projects.

The course is straightforward – if you do the work, you will do well.

Your mark is 80% perspiration and 20% inspiration.

Why are you here?

- A)** It is a required course for the COSC/DATA major.
- B)** This is an optional elective for my program.
- C)** Engineering Major taking a minor in Computer Science.
- D)** Taking a minor in Data Science.
- E)** Other

What Topic are You Most Interested In?

- A)** What is a database and how do you use them?
- B)** Querying using SQL
- C)** Designing databases
- D)** Using databases with programs (stand-alone, web, mobile)
- E)** Using databases for data analysis and visualization

Database Survey Question

Question: Have you used any of these database systems?

- A)** MySQL
- B)** Microsoft Access or SQL Server
- C)** PostgreSQL
- D)** Used more than two different databases
- E)** Used no databases

What Grade are You Expecting to Get?

A) A

B) B

C) C

D) D

F) F

The Essence of the Course

Essence of the course is to appreciate that:

Database systems are a powerful and widely used way to store and manipulate *persistent* information. You will learn the skills to exploit the full power of database systems.

The skills you will acquire are in high demand for many software development jobs. Database skills make you more marketable and allow you to construct more sophisticated systems.

- Note: This is a course on how to use/program with databases. It is a very applied course with specific skills.
- If you want to learn how to build database systems and what is “inside the box”, that is the subject of COSC 404!



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